IX. Consistent linearization

1. Directional derivative
2. Linearization concept
3. Continuous and discrete examples
4. Linearization of the internal virtual work
5. Geometric stiffness matrix
6. Material stiffness matrix

Directional derivative, Linearization

Consider an arbitrary nonlinear function $\boldsymbol{f}(\boldsymbol{x})$ mapping $\boldsymbol{x} \in \mathbb{R}^n$ to $\mathbb{R}^m$

Directional derivative of $\boldsymbol{f}(\boldsymbol{x})$ at $\bar{\boldsymbol{x}}$ in the direction $\Delta\boldsymbol{x}$

$$\mathrm{D}\boldsymbol{f}(\bar{\boldsymbol{x}}, \Delta\boldsymbol{x}) = \mathrm{D}\boldsymbol{f}(\bar{\boldsymbol{x}})[\Delta\boldsymbol{x}] := \lim_{\eta \to 0} \frac{\boldsymbol{f}(\bar{\boldsymbol{x}} + \eta\Delta\boldsymbol{x}) - \boldsymbol{f}(\bar{\boldsymbol{x}})}{\eta} = \left[\frac{\mathrm{d}}{\mathrm{d}\eta}\boldsymbol{f}(\bar{\boldsymbol{x}} + \eta\,\Delta\boldsymbol{x})\right]_{\eta=0}$$

$\rightarrow$ Also known as the Gataux differential of $\boldsymbol{f}(\boldsymbol{x})$

If $\mathrm{D}\boldsymbol{f}(\bar{\boldsymbol{x}}, \Delta\boldsymbol{x})$ is linear in $\Delta\boldsymbol{x}$ we can write

$$\mathrm{D}\boldsymbol{f}(\bar{\boldsymbol{x}}, \Delta\boldsymbol{x}) = \left.\frac{\partial\boldsymbol{f}}{\partial\boldsymbol{x}}\right|_{\bar{\boldsymbol{x}}} \cdot \Delta\boldsymbol{x}$$

$\rightarrow$ Also known as the Frechet differential of $\boldsymbol{f}(\boldsymbol{x})$

$\rightarrow$ This defines the derivative of $\boldsymbol{f}(\boldsymbol{x})$

Linearization of $\boldsymbol{f}(\boldsymbol{x})$ at $\bar{\boldsymbol{x}}$ in the direction $\Delta\boldsymbol{x}$

$$\mathrm{L}[\boldsymbol{f}, \Delta\boldsymbol{x}]_{\bar{\boldsymbol{x}}} = \boldsymbol{f}(\bar{\boldsymbol{x}}) + \mathrm{D}\boldsymbol{f}(\bar{\boldsymbol{x}}, \Delta\boldsymbol{x})$$

Recall Taylor expansion

Examples: $\mathrm{L}[\boldsymbol{F}, \Delta\boldsymbol{x}]_{\bar{\boldsymbol{x}}} =$

$\mathrm{L}[J, \Delta\boldsymbol{x}]_{\bar{\boldsymbol{x}}} =$

Linearization of the weak form

Linearization of the weak form

$$G(\boldsymbol{u}) = G_{\text{int}}(\boldsymbol{u}) + G_{\text{ext}}$$

$$\mathrm{L}\big[G, \Delta\boldsymbol{u}\big]_{\bar{\boldsymbol{u}}} = G(\bar{\boldsymbol{u}}) + \mathrm{D}G_{\text{int}}(\bar{\boldsymbol{u}}, \Delta\boldsymbol{u})$$

At the element level

$$G^e_{\text{int}}(\boldsymbol{u}) = \int_{\Omega^e_0} \delta\boldsymbol{F} : \boldsymbol{P}\,\mathrm{d}V \ , \quad \delta\boldsymbol{F} = \mathrm{Grad}\,(\delta\boldsymbol{u}) = \frac{\partial\,\delta\boldsymbol{u}}{\partial\boldsymbol{X}} \ , \quad \boldsymbol{P} = \boldsymbol{F}\boldsymbol{S}$$

$$\begin{aligned}
\mathrm{D}G^e_{\text{int}}(\bar{\boldsymbol{u}}, \Delta\boldsymbol{u}) &= \int_{\Omega^e_0} \delta\boldsymbol{F} : \big(\mathrm{D}\boldsymbol{F}(\bar{\boldsymbol{u}}, \Delta\boldsymbol{u})\,\boldsymbol{S} + \boldsymbol{F}\,\mathrm{D}\boldsymbol{S}(\bar{\boldsymbol{u}}, \Delta\boldsymbol{u})\big)\,\mathrm{d}V \\
&= \underbrace{\int_{\Omega^e_0} \delta\boldsymbol{F} : \mathrm{D}\boldsymbol{F}(\bar{\boldsymbol{u}}, \Delta\boldsymbol{u})\,\boldsymbol{S}\,\mathrm{d}V}_{\substack{\text{geometrical part} \\ \mathrm{D}G^e_{\text{geo}}(\bar{\boldsymbol{u}}, \Delta\boldsymbol{u})}} + \underbrace{\int_{\Omega^e_0} \delta\boldsymbol{F} : \boldsymbol{F}\,\mathrm{D}\boldsymbol{S}(\bar{\boldsymbol{u}}, \Delta\boldsymbol{u})\,\mathrm{d}V}_{\substack{\text{material part} \\ \mathrm{D}G^e_{\text{mat}}(\bar{\boldsymbol{u}}, \Delta\boldsymbol{u})}}
\end{aligned}$$

Geometrical part

Continuous setting

$$\mathrm{D}G^e_{\mathrm{geo}}(\bar{\boldsymbol{u}}, \Delta\boldsymbol{u}) = \int_{\Omega^e_0} \frac{\partial\,\delta\boldsymbol{u}}{\partial\boldsymbol{X}} : \mathrm{D}\boldsymbol{F}(\bar{\boldsymbol{u}}, \Delta\boldsymbol{u})\,\boldsymbol{S}\,\mathrm{d}V = \int_{\Omega^e_0} \frac{\partial\,\delta\boldsymbol{u}}{\partial\boldsymbol{X}} : \frac{\partial\,\Delta\boldsymbol{u}}{\partial\boldsymbol{X}}\boldsymbol{S}\,\mathrm{d}V$$

$$\boldsymbol{S} = J\,\boldsymbol{F}^{-1}\boldsymbol{\sigma}\,\boldsymbol{F}^{-T}$$

$$= \int_{\Omega^e} \frac{\partial\,\delta\boldsymbol{u}}{\partial\boldsymbol{X}}\boldsymbol{F}^{-1} : \frac{\partial\,\Delta\boldsymbol{u}}{\partial\boldsymbol{X}}\boldsymbol{F}^{-1}\boldsymbol{\sigma}\,\mathrm{d}v$$

$$= \int_{\Omega^e} \frac{\partial\,\delta\boldsymbol{u}}{\partial\boldsymbol{x}} : \frac{\partial\,\Delta\boldsymbol{u}}{\partial\boldsymbol{x}}\boldsymbol{\sigma}\,\mathrm{d}v = \int_{\Omega^e} \delta u_{i,j}\,\sigma_{jk}\,\Delta u_{i,k}\,\mathrm{d}v$$

$$= \int_{\Omega^e} \delta\boldsymbol{u}^T_{,i}\,\sigma_{ij}\,\Delta\boldsymbol{u}_{,j}\,\mathrm{d}v = \int_{\Omega^e_0} \delta\boldsymbol{u}^T_{,A}\,S_{AB}\,\Delta\boldsymbol{u}_{,B}\,\mathrm{d}V$$

$\rightarrow$ symmetric w.r.t. $\delta\boldsymbol{u}$ and $\Delta\boldsymbol{u}$

Discrete setting

$$\delta\boldsymbol{u} = \mathbf{N}\,\mathbf{v}_e$$
$$\Delta\boldsymbol{u} = \mathbf{N}\,\Delta\mathbf{u}_e$$

$$\mathrm{D}G^e_{\mathrm{geo}}(\bar{\boldsymbol{u}}, \Delta\boldsymbol{u}) = \mathbf{v}_e^T \int_{\Omega^e_0} \mathbf{N}^T_{,I}\,S_{IJ}\,\mathbf{N}_{,J}\,\mathrm{d}V\;\Delta\mathbf{u}_e$$

$$= \mathbf{v}_e^T \underbrace{\int_{\Omega^e} \mathbf{N}^T_{,i}\,\sigma_{ij}\,\mathbf{N}_{,j}\,\mathrm{d}v}\;\Delta\mathbf{u}_e$$

$$= \mathbf{v}_e^T \quad \mathbf{K}^e_{\mathrm{geo}} \quad \Delta\mathbf{u}_e$$

$$\rightarrow\quad (\mathbf{K}^e_{\mathrm{geo}})^T = \mathbf{K}^e_{\mathrm{geo}}$$

Material part

Material law (considered differentiable)

$$\boldsymbol{S} = \boldsymbol{S}(\boldsymbol{E}) \ , \quad \boldsymbol{E} = \tfrac{1}{2}\big(\boldsymbol{F}^T\boldsymbol{F} - \boldsymbol{I}\big)$$

$$\mathrm{D}\boldsymbol{S} = \frac{\partial \boldsymbol{S}}{\partial \boldsymbol{E}} : \Delta\boldsymbol{E} \ , \quad \Delta\boldsymbol{E} = \frac{1}{2}\big(\Delta\boldsymbol{F}^T\boldsymbol{F} + \boldsymbol{F}^T\Delta\boldsymbol{F}\big)$$

Continuous setting

$$\begin{aligned}
\mathrm{D}G^e_{\mathrm{mat}}(\bar{\boldsymbol{u}}, \Delta\boldsymbol{u}) &= \int_{\Omega^e_0} \delta\boldsymbol{F} : \boldsymbol{F}\,\mathrm{D}\boldsymbol{S}(\bar{\boldsymbol{u}}, \Delta\boldsymbol{u})\,\mathrm{d}V \\
&= \int_{\Omega^e_0} \frac{\partial\,\delta\boldsymbol{u}}{\partial\boldsymbol{X}} : \boldsymbol{F}\,\frac{\partial\boldsymbol{S}}{\partial\boldsymbol{E}} : \frac{1}{2}\Big[\Big(\frac{\partial\,\Delta\boldsymbol{u}}{\partial\boldsymbol{X}}\Big)^T\boldsymbol{F} + \boldsymbol{F}^T\frac{\partial\,\Delta\boldsymbol{u}}{\partial\boldsymbol{X}}\Big]\,\mathrm{d}V \\
&= \int_{\Omega^e_0} \frac{\partial\,\delta\boldsymbol{u}}{\partial\boldsymbol{x}}\boldsymbol{F} : \boldsymbol{F}\,\frac{\partial\boldsymbol{S}}{\partial\boldsymbol{E}} : \frac{1}{2}\Big[\boldsymbol{F}^T\Big(\frac{\partial\,\Delta\boldsymbol{u}}{\partial\boldsymbol{x}}\Big)^T\boldsymbol{F} + \boldsymbol{F}^T\frac{\partial\,\Delta\boldsymbol{u}}{\partial\boldsymbol{x}}\boldsymbol{F}\Big]\,\mathrm{d}V \\
&= \int_{\Omega^e_0} \boldsymbol{F}^T\frac{\partial\,\delta\boldsymbol{u}}{\partial\boldsymbol{x}}\boldsymbol{F} : \frac{\partial\boldsymbol{S}}{\partial\boldsymbol{E}} : \boldsymbol{F}^T\Big(\frac{\partial\,\Delta\boldsymbol{u}}{\partial\boldsymbol{x}}\Big)^s\boldsymbol{F}\,\mathrm{d}V
\end{aligned}$$

Note

$$\frac{\partial\boldsymbol{S}}{\partial\boldsymbol{E}} = \frac{\partial\boldsymbol{S}}{\partial\boldsymbol{\sigma}} : \frac{\partial\boldsymbol{\sigma}}{\partial\boldsymbol{e}} : \frac{\partial\boldsymbol{e}}{\partial\boldsymbol{E}}$$

$$\boldsymbol{S} = J\boldsymbol{F}^{-1}\boldsymbol{\sigma}\boldsymbol{F}^{-T} \quad \rightarrow \quad \frac{\partial S_{AB}}{\partial\sigma_{ij}} = J\,F^{-1}_{Ai}\,F^{-1}_{Bj}$$

$$\boldsymbol{e} = \boldsymbol{F}^{-1}\boldsymbol{E}\boldsymbol{F}^{-T} \quad \rightarrow \quad \frac{\partial e_{ij}}{\partial E_{AB}} = F^{-1}_{Ai}\,F^{-1}_{Bj}$$

Material part (continued)

Continuous setting

$$\mathrm{D}G^e_{\mathrm{mat}}(\bar{\boldsymbol{u}}, \Delta\boldsymbol{u}) = \int_{\Omega^e_0} \boldsymbol{F}^T \frac{\partial\,\delta\boldsymbol{u}}{\partial\boldsymbol{x}} \boldsymbol{F} : \frac{\partial\boldsymbol{S}}{\partial\boldsymbol{\sigma}} : \frac{\partial\boldsymbol{\sigma}}{\partial\boldsymbol{e}} : \frac{\partial\boldsymbol{e}}{\partial\boldsymbol{E}} : \boldsymbol{F}^T \left(\frac{\partial\,\Delta\boldsymbol{u}}{\partial\boldsymbol{x}}\right)^s \boldsymbol{F}\,\mathrm{d}V$$

$$= \int_{\Omega^e} \underbrace{\frac{1}{J}\boldsymbol{F}^T \frac{\partial\,\delta\boldsymbol{u}}{\partial\boldsymbol{x}} \boldsymbol{F} : \frac{\partial\boldsymbol{S}}{\partial\boldsymbol{\sigma}}} : \frac{\partial\boldsymbol{\sigma}}{\partial\boldsymbol{e}} : \underbrace{\frac{\partial\boldsymbol{e}}{\partial\boldsymbol{E}} : \boldsymbol{F}^T \left(\frac{\partial\,\Delta\boldsymbol{u}}{\partial\boldsymbol{x}}\right)^s \boldsymbol{F}}\,\mathrm{d}v$$

$$= \int_{\Omega^e} \frac{\partial\,\delta\boldsymbol{u}}{\partial\boldsymbol{x}} : \frac{\partial\boldsymbol{\sigma}}{\partial\boldsymbol{e}} : \left(\frac{\partial\,\Delta\boldsymbol{u}}{\partial\boldsymbol{x}}\right)^s \mathrm{d}v$$

$$= \int_{\Omega^e} \left(\frac{\partial\,\delta\boldsymbol{u}}{\partial\boldsymbol{x}}\right)^s : \mathfrak{c} : \left(\frac{\partial\,\Delta\boldsymbol{u}}{\partial\boldsymbol{x}}\right)^s \mathrm{d}v\,, \quad \mathfrak{c} := \frac{\partial\boldsymbol{\sigma}}{\partial\boldsymbol{e}}$$

Discrete setting

$$\mathrm{grad}\,(\delta\boldsymbol{u})^s = \mathbf{B}_{\mathrm{UL}}\,\mathbf{v}_e$$

$$\mathrm{grad}\,(\Delta\boldsymbol{u})^s = \mathbf{B}_{\mathrm{UL}}\,\Delta\mathbf{u}_e$$

$$\mathrm{D}G^e_{\mathrm{mat}}(\bar{\boldsymbol{u}}, \Delta\boldsymbol{u}) = \mathbf{v}_e^T \underbrace{\int_{\Omega^e} \mathbf{B}_{\mathrm{UL}}^T\,\mathfrak{c}_{\mathrm{v}}\,\mathbf{B}_{\mathrm{UL}}\,\mathrm{d}v} \Delta\mathbf{u}_e = \mathbf{v}_e^T \quad \mathbf{K}^e_{\mathrm{mat}} \quad \Delta\mathbf{u}_e$$

→ Linearization of the weak form

$$\mathrm{L}\big[G, \Delta\boldsymbol{u}\big]_{\bar{\boldsymbol{u}}} = G(\bar{\boldsymbol{u}}) + \mathrm{D}G_{\mathrm{int}}(\bar{\boldsymbol{u}}, \Delta\boldsymbol{u})$$

$$= \mathbf{v}^T\,\mathbf{f} + \mathbf{v}^T\,\mathbf{K}_{\mathrm{int}}\,\Delta\mathbf{u} = \mathbf{v}^T\big(\mathbf{f} + \mathbf{K}_{\mathrm{int}}\,\Delta\mathbf{u}\big)\,, \qquad \mathbf{K}_{\mathrm{int}} = \mathbf{K}_{\mathrm{geo}} + \mathbf{K}_{\mathrm{mat}}$$